

The Matrix Multiplication Exponent Race

From Strassen (1969) to a 2026 Machine-Assisted Bound – and Why Big-O Still Hides the Truth

Week 6 Supplement – TOAE209/TOAH209: Design and Analysis of Algorithms

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Outline

A question the lecture already half-answers

- This week's Master Theorem example for “grade-school multiply (D&C)” is a cautionary tale: splitting an n -digit multiplication into 4 half-size sub-multiplications gives $T(n) = 4T(n/2) + \Theta(n)$, and Case 1 says this is $\Theta(n^2)$ – **no better than the naive algorithm**. Karatsuba's insight was to cut 4 sub-multiplications down to 3.
- The obvious next question: **multiplying two $n \times n$ matrices** the “grade-school” way (triple nested loop) costs $\Theta(n^3)$. If we split each matrix into four $(n/2) \times (n/2)$ blocks, the naive block recursion also needs 8 sub-multiplications: $T(n) = 8T(n/2) + \Theta(n^2)$, and $\log_2 8 = 3$, so again **no improvement** – exactly the same trap.
- Is there a “Karatsuba trick” for matrices too – a way to shave sub-multiplications and beat n^3 ? Yes – and the race to find the *best possible* such trick has been running for over 55 years, with a new record set as recently as this month.

Setup: block recursion

- Split each $n \times n$ matrix into four $(n/2) \times (n/2)$ blocks: $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$.
- The defining identity of block matrix multiplication needs **8** products of half-size blocks (e.g. $C_{11} = A_{11}B_{11} + A_{12}B_{21}$, and similarly for C_{12} , C_{21} , C_{22}) plus $\Theta(n^2)$ additions to combine them.
- $T(n) = 8T(n/2) + \Theta(n^2)$, $\log_2 8 = 3 = d + 1 \dots$ by the Master Theorem (Case 1, $\log_2 8 = 3 > 2$): $\Theta(n^{\log_2 8}) = \Theta(n^3)$. Same as naive.

Strassen's trick: 7 products instead of 8

- Volker Strassen (1969) found **seven** cleverly-chosen linear combinations of the blocks whose products (plus $\Theta(n^2)$ extra additions/subtractions) can be recombined into all four quadrants of C – one fewer multiplication than the obvious approach, at the cost of more (but still linear-in- n^2) addition work.
- New recurrence: $T(n) = 7T(n/2) + \Theta(n^2)$.
- Master Theorem: $\log_2 7 \approx 2.807 > 2$, so Case 1 gives $T(n) = \Theta(n^{\log_2 7}) \approx \Theta(n^{2.807})$.

Why this was a big deal

For over a century, $\Theta(n^3)$ was assumed to be the best possible. Strassen's algorithm was the **first proof that matrix multiplication is not cubic** – exactly the same kind of shock Karatsuba's algorithm gave integer multiplication, six years earlier.

The exponent ω : how low can it go?

Define ω as the infimum exponent such that $n \times n$ matrices can be multiplied in $O(n^{\omega+\epsilon})$ time for every $\epsilon > 0$. Strassen showed $\omega \leq \log_2 7 \approx 2.807$; the true value of ω is a major open problem (it is known $\omega \geq 2$, trivially). Since 1969, a long sequence of results has driven the upper bound down:

Year	Result	Bound on ω
1969	Strassen	2.807
1990	Coppersmith & Winograd	2.376
2023	Duan, Wu & Zhou (FOCS)	2.371866
2024	Vassilevska Williams, Xu, Xu & Zhou (SODA)	2.371552
2025	Alman, Duan, Vassilevska Williams, Xu, Xu & Zhou (SODA)	2.371339
2026	modern-optimization + AlphaEvolve search (arXiv 2608.16884)	2.371177

Every one of these (2023 onward) is a refinement of the same underlying technique – the *laser method* applied to powers of a combinatorial object (the Coppersmith–Winograd tensor) – pushed further each time by a sharper loss analysis.

What actually moved between 2023 and 2026

- **2023 (Duan, Wu, Zhou, FOCS):** identified that earlier analyses of powers of the Coppersmith–Winograd tensor suffered from an avoidable “combination loss,” and fixed it – $\omega < 2.371866$.
- **2024 (Vassilevska Williams, Xu, Xu, Zhou, SODA, “From Alpha to Omega”):** a refined laser-method analysis, also improving the related *rectangular* exponent $\alpha - \omega \leq 2.371552$.
- **2025 (Alman, Duan, Vassilevska Williams, Xu, Xu, Zhou, SODA, “More Asymmetry Yields Faster Matrix Multiplication”):** deliberately *unbalancing* the analysis (treating the tensor’s three dimensions asymmetrically) squeezes out more – $\omega < 2.371339$.
- **2026 (arXiv 2608.16884):** the same optimization problem these papers solve by hand is reformulated and handed to a machine-learning-guided search (Google DeepMind’s AlphaEvolve) over a larger search space – $\omega < 2.371177$.

The pattern

Four papers, three years, the fifth decimal place – this is what “chasing a constant” looks like at the frontier of theoretical computer science.

Why your laptop's BLAS library still uses Strassen, not $\omega < 2.372$

- Strassen's algorithm *is* used in practice, but only above a **crossover point**: the constant-factor overhead of its extra additions only pays off once matrices are large enough (empirically, roughly n in the tens to low hundreds, and highly hardware-dependent) – below that, plain $\Theta(n^3)$ wins.
- Every algorithm since Coppersmith–Winograd (1990) is what Richard Lipton called a **“galactic algorithm”**: asymptotically faster, but with hidden constants (and exponents on those constants) so large that the crossover point exceeds any matrix size that could ever exist in a real computation.
- This is a direct, concrete instance of the warning from Week 2's supplement: “textbook algorithm” does not mean “final answer” – and here, “asymptotically optimal” does not even mean “ever useful.”

Even a galactic bound has real consequences

- Many other problems' *best known* running times are stated *in terms of* ω – e.g. Boolean matrix multiplication (used for transitive closure and some graph reachability algorithms), certain all-pairs-shortest-path methods, and context-free grammar parsing all inherit an $O(n^\omega)$ -flavoured bound.
- Every time ω 's upper bound drops, *all of those* bounds improve too, even though nobody re-proves each one by hand – the improvement propagates through the literature automatically.
- So while you will never see a $\Theta(n^{2.371})$ matrix multiply run on real hardware, the *pursuit* of ω is not idle: it is calibrating exactly how much headroom is theoretically left in dozens of other algorithms you might build later in this course and beyond.

Summary

- 1 The same “count the sub-multiplications, apply the Master Theorem” analysis this week uses for integer multiplication applies verbatim to matrix multiplication – 8 sub-products gives $\Theta(n^3)$ (no gain), Strassen's 7 gives $\Theta(n^{2.807})$ (a real gain).
- 2 The exponent ω has been pushed from 2.807 (1969) to 2.371177 (Aug 2026) through a sequence of genuinely real, citable papers – most of them from the last three years, the newest from this very month.
- 3 Every algorithm below Coppersmith–Winograd's 1990 bound is “galactic”: correct, asymptotically superior, and never actually run – a sharp reminder that Big- O describes a *limit*, not a *promise about your input size*.

Looking ahead

Next week (Divide & Conquer II) revisits integer multiplication one level deeper: Harvey and van der Hoeven's true $O(n \log n)$ algorithm (2019–2021) – the actual endpoint of the road Karatsuba started Week 1 on.